

Aditya Tayal

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Final-year CSE undergraduate at IIT Mandi who ships production software end to end. Solo-built PlanMyDegree, now used by **1,021 IIT Mandi students**, plus faculty-leave, exam and lab systems for institute offices. Four startup internships across travel-tech payments, solar-energy ML and counter-drone systems; **All India Rank 1** in the SAND 2025 (ICASSP) challenge; elected **UG Academic Affairs Secretary** for ~3,500 undergraduates.

EDUCATION

Indian Institute of Technology Mandi

B.Tech in Computer Science & Engineering | CGPA: 8.80

Mandi, Himachal Pradesh

Aug 2023 – May 2027 (expected)

Relevant Coursework: Machine Learning, Deep Learning (CS-671), Advanced Machine Learning / Generative AI (CS-683), Data Science I & II, Data Handling & Visualization, Database Systems, Design of Algorithms, Automata Theory, Mathematical Foundations of CS, Operating Systems, Computer Networks, Information Security & Privacy, Optimization, Paradigms of Programming

Ramakrishna Mission

Higher Secondary Education in PCM with Computer Science | 89.40%

Gwalior, Madhya Pradesh

Aug 2020 – Jun 2022

Wendy School Junior College

ICSE Secondary Education | 98.20%

Gwalior, Madhya Pradesh

Apr 2007 – Jul 2020

EXPERIENCE

Major Technical Project (MTP), American Express

2026 – 2027 (in progress)

- Final-year Major Technical Project with American Express.

Software Developer Intern, Koha Consultants (InstaFarms & MagoStays)

Apr 2026 – Present

- Ship full-stack features across **5 production codebases** of a villa & farmstay booking platform running two brands: a TypeScript Turborepo backend (Hono + zod-openapi, Drizzle, PostgreSQL), two Next.js 15 customer sites, an admin panel and an Expo React Native host app; **485 commits**.
- Built the owner-payments stack end to end: Razorpay Route onboarding & direct transfers, host wallet, RazorpayX payouts, penny-drop bank verification with name match, and owner KYC.
- Built the invoice engine: exactly-once invoice serial numbering, maker-checker corrections, a Zoho Books outbox, and GST/TCS/TDS logic across **4 owner agreement models**.
- Integrated channel partner Elivaas (catalog sync, Razorpay checkout, SEO city/state pages) and cut availability-calendar API calls from **60 to 2**.
- Implemented OTA booking cancellation from the PRD with payout accounts per brand and per OTA (Airbnb, Booking.com, MakeMyTrip, Agoda, Goibibo); added a per-room Resort property type across all 5 apps covering **24 specified edge cases**.
- Revamped the monorepo with a self-documenting OpenAPI/Swagger layer; wrote **50 SQL migrations** and **65 test files**, plus Playwright end-to-end checks and a Docker + Caddy staging pipeline.

AI/ML Engineer Intern, Wattcycle Sustainable Solutions (WattCycle.ai)

Dec 2025 – Jun 2026

- Worked across full stack and ML for 7 months on WattCycle Atlas, a solar-plant monitoring platform: data ingestion, API, forecasting, dashboard and deployment.
- Built fault detection & energy forecasting with Isolation Forest, LSTM (TensorFlow/Keras) and a Prophet + XGBoost ensemble across **13 forecast horizons**, on **50+ engineered time-series features**.
- Set up MLOps with MLflow and automated retraining; served real-time inference via FastAPI with Kafka event streaming.
- Made the data trustworthy: idempotent, retry-safe writes, provenance labels on every metric, labelled uncertainty bands and change-aware CI.

Full Stack Developer Intern, Garudakshak Technologies (Counter-UAV)

Oct 2025 – Apr 2026

- Took a Windows desktop app (Tauri v2 + Rust, React, TypeScript) from **zero to MVP** for controlling and monitoring a counter-drone system: sensor feeds, cameras and detection pipelines.
- Built a Go API gateway with **30+ endpoints** and Keycloak SSO, and a real-time WebSocket dashboard streaming optical, acoustic and RF sensor data.
- Built a Raspberry Pi sensor & camera service that recovers from device disconnects and restarts on its own; worked with hardware and AI teams to wire real-time ML threat classification into the operator UI.

Software Developer Intern (Lead Developer), Strive Partners (Hunt)

Sep 2025 – Jun 2026

- Led backend development: Django + PostgreSQL services on AWS (EC2, RDS, S3), later moved to DigitalOcean, with Redis caching, SMTP notifications and CORS for a Vercel-hosted frontend.
- Built secure REST APIs for the React frontend and owned **3 product modules** end to end; added Redis caching on hot queries to cut API latency.

Software for IIT Mandi**PlanMyDegree: Academic Planner for IIT Mandi (Solo)**[Live](#) [Code](#)*Next.js 15, React 19, TypeScript, Prisma, PostgreSQL, NextAuth, Vercel, PWA**Jan 2026 – Present*

- Solo-built a full-stack academic planner now used by **1,021 students (67% of those eligible)** across 4 batches; **150+ sign-ups in 36 hours** from one WhatsApp message, no ads.
- Tracks **17,555 enrollments** and **51,915 credits**; students built **347 pre-registration plans**; covers 1,427 courses, 13 branch curricula, 15 minors and 5 German exchange universities.
- Pre-registration with seat caps, mutual exclusion of equivalent courses and declarations synced to Google Sheets via an authenticated webhook; transcript OCR import, timetable clash detection, calendar export.
- **30 data models, ~65 API routes, 880+ commits**; average feedback rating 4.9 from 27 reviews.

Faculty Leave Portal: Dean of Faculty Affairs Office (Solo)[Pilot](#)*Fastify, Prisma, React, Vite, Tailwind, SQLite, Docker, Raspberry Pi**Sep 2026 – Present*

- Built Phase 1 of faculty leave management from the institute's PRD: rule-checked applications, an auditable approval chain and establishment records; traced **324 PRD requirements** with **321 automated tests**.
- No stored balances: leave is derived on every read from an append-only ledger; DB triggers block edits on ledger and audit tables; two-phase HOLD → DEBIT posting; bcrypt-hashed OTP, CSRF, magic-byte upload checks.

TestPlatform: Secure Online Examination System (Solo)*Next.js, Express, Socket.IO, PostgreSQL, Redis, nginx, pm2**Aug 2026 – Present*

- Student, teacher and admin portals with multi-tenant organisations; server-authoritative exam timers in Redis with Socket.IO live rooms; auto-grading for MCQ, multi-select and numeric-with-tolerance answers.
- Passwordless OTP over email and WhatsApp; hardened after a self-audit: pinned JWT algorithm, row-locked final submit (SELECT ... FOR UPDATE), IDOR fixes, 2 CVE patches.

Exodia 2026 Ticketing and Gate Management System (Solo)[Code](#)*Next.js 15, tRPC, Prisma, Neon Postgres, NextAuth, Nodemailer**Mar 2026 – Apr 2026*

- Built QR ticketing, email delivery and gate scanning for the fest's Pronite: **300+ QR tickets** emailed, **200 attendees** admitted, **0 duplicate entries**.
- Atomic single-use QR claims so concurrent gate scans cannot double-admit; **blocked 26 reuse attempts**; shipped **13 commits live** on event day with zero downtime.

Faculty Application Portal, IIT Mandi[Live](#)*Django, SQL, HTML/CSS**Dec 2024 – Feb 2025*

- Replaced Google Forms with a secure Django portal processing **200+ faculty applications** annually; admin dashboard to filter, track and export; automated CV generation & emails, saving **20+ admin hours/month**.

Atal Tinkering Lab Tools: TinkerBench, Simulation Lab, NCERT Sim (Solo)*JavaScript, Canvas, nginx, Docker**Sep 2026 – Oct 2026*

- Browser electronics bench solving circuits by nodal analysis every frame (diodes, transistors, a working 555 timer) and running Arduino C++ translated to JS; **23 Class 9–12 experiments** and **26 NCERT physics practicals**.

IIT Mandi LaTeX: Institute-hosted Overleaf Alternative (Lead)*Overleaf CE, Node.js, Docker, CI**Sep 2026 – Present*

- Extended Overleaf CE for ~3,500 users: FCFS compile queue, 50-user cap with waiting room, 500 MB quotas, nightly off-host backups; costed at **INR 1.26 lakh one-time vs INR 11.29 lakh/yr**.

Machine Learning & Research**Camouflaged Object Detection (Paid Research Project)***PyTorch, mmdetection, DINOv2, LoRA, SigLIP-2, SAM2**Apr 2026 – Jul 2026*

- Adapted PAIS (ICCV 2023) to camouflaged objects with 10% labels: COD10K AP **29.7 → 33.0, +7.8 AP** over the supervised baseline; NC4K AP 46.9.
- Designed LGCAD, a DINOv2 + LoRA decoder with SigLIP-2 text-guided queries (22 variants, 4 datasets): S-measure **0.847** (COD10K), **0.865** (NC4K).
- Reproduced training-free UCOD-DPL (CVPR 2025) with two-pass SAM2 refinement and 25-mask soft voting: S-measure **0.8407 vs paper's 0.835**; improved GNet on **42/42** dataset-metric checks over 71 experiment phases.

Domain Unlearning in CLIP (CS-683, Team of 4)[Code](#)*PyTorch, CLIP ViT-B/16, Prompt Learning, SCRUB**Jan 2026 – May 2026*

- Re-implemented ADU and beat the published Office-Home H-score (**69.96 → 72.10**) by fixing an N:1 loss imbalance between kept and forgotten domains.
- Adapted SCRUB from sample- to domain-level unlearning: **83.6 H-score, +13.6 points** over the paper; 82.9 average across 6 two-domain pairs; IEEE-format report.

Vehicle Counting with Classical CV (Vehant Challenge, Solo)*OpenCV, NumPy**Jan 2026*

- 9-stage pipeline in 48 hours with deep learning banned (GMM background subtraction, motion ROI, centroid tracking, track merging, gate-line counting, night mode): **MAE 16.4, rank 10** against teams.

MahaDrishti AI: Satellite Crop Verification (Pune Agri Hackathon Finalist)

Sentinel-1/2, Prithvi-EO 2.0, Random Forest, FastAPI, Gemini

May 2026

- Verifies farmers' declared crops per land-record plot from satellite imagery with Marathi/English reports; RF 5-fold CV **81.86%**, mixture-of-experts **83.21%**; 194 plots per village in under 2 minutes.

Product Price Prediction (Amazon ML Challenge 2025, Team of 3)

LightGBM, TF-IDF, Python

Oct 2025

- TF-IDF + 18 regex features into a 21-model LightGBM ensemble (7-fold $\times$ 3 seeds) with calibration: SMAPE **54.38%** $\rightarrow$ **32.33%** out-of-fold.

Deepfake Detection using Vision Transformers (CS-671, Team of 6)

CLIP ViT-L/14, ViT-H/14, DINOv2, SVM (cuML)

Feb 2025 – May 2025

- Frozen ViT features + GPU RBF-SVM evaluated on **19 generators** and **80,353** test images: **89.4%** accuracy weighted by test-set size.

CaptureSmart: AI Camera Assistant for Android (Team of 6)

Kotlin, CameraX, TensorFlow, MobileNetV2, Flask

[Code](#)

May 2025

- MobileNetV2 blur scorer (MAE 6.7, Pearson 0.89 on 24,000 images) plus ISO/shutter recommenders (shutter R^2 0.912); secured **3rd** place among **60+** teams.

Applied AI & Hackathon Builds

FinFabric: Applied AI for Banking (IDBI Innovate 2026, Team Lead)

JavaScript, Vercel

[Code](#)

Jul 2026

- Led a 5-product live suite: MSME health card, mutual-fund SIP companion with live NAVs, lead scoring, default early-warning model and a fuzzy-matching reconciliation copilot.

Sahayak: Conversational AI for a Crime Database (KSP Datathon 2026)

Qwen 2.5 Coder 7B, Zoho Catalyst, FastAPI, React

[Code](#)

Jun 2026

- English/Kannada questions to SQL over 3,500 FIRs with a read-only SQL guard; every answer shows its SQL and source rows; IPC $\leftrightarrow$ BNS legal-code crosswalk.

SevaVoice (HACKHAZARDS '26) and DeepMatch (India Runs)

Sarvam AI, React, TypeScript, MiniLM, Python

[Live Code](#)

Jun 2026

- SevaVoice turns public-service notices into verified action plans in **14 Indian languages** (voice in 11), with 63 Playwright checks.
- DeepMatch ranks candidates in 3 stages (rules 100K $\rightarrow$ 5K, MiniLM scoring, signal fusion) with 6 fake-profile heuristics in $\sim$ 75 s on CPU.

DyslexoFly: Accessible Reading Platform (Code for Bharat S2)

Next.js 15, Flask, DistilBART, mT5, edge-tts

[Live Code](#)

Jun 2025 – Jul 2025

- Converts PDF/DOCX/TXT into dyslexia-friendly reading with OpenDyslexic, English and Hindi summaries, and audio in 7 neural voices.

Adobe India Hackathon 2025: PDF Intelligence (Team Leader)

Python, PyMuPDF, MiniLM, Docker

[Code](#)

Jul 2025

- Reached Round 2 (**top 1.5%** of 262,120 teams): multilingual PDF outline extraction in an offline CPU-only Docker image and persona-aware ranking of 7 PDFs in 30–45 s.

More builds:

- **FlowAI** (Vibe2Ship 2026): Gemini multi-agent productivity app on Next.js + Firebase. **PurrPedia** (HackTheKitty 2026): Next.js 16 + Temporal workflows; fixed all 5 security-audit findings.
- **TheBudgetBuddy** (FrostHack 2025): LangChain agent with Groq Llama3-70B, Cohere rerank and FAISS. **CareerAutofill AI**: agents + Chrome extension; LoRA on distilgpt2 cut perplexity 35.7 $\rightarrow$ 23.2.
- **CyberShield** (PSBs Hackathon, IIT Hyderabad): mule-account detection on 9,082 accounts (0.89% base rate). **Vidyut AI** (ET AI Hackathon 2.0): citation-grounded Q&A over safety documents.
- **Face EngagementCV**: dlib + MediaPipe face ID and engagement scoring from CCTV. **PSCDL 2026**: classical-CV scene change detection. **TheAttendioBot**: Telegram attendance planner.
- **SAND 2025**: ALS severity from voice (macro-F1 0.5430). **ISTP**: solar PV waste policy study with field visit. **Autonomous ROS robot**: ESP32 on a custom 2-layer PCB.
- Also: malware scanner (FastAPI + YARA), SecureInvest-AI (SEBI GFF 2025), PodSense (ABB 2026), UGAC website, GCS website, Neufin, PRAYAS ML labs, LaTeX lecture notes.

POSITIONS OF RESPONSIBILITY

UG Academic Affairs Secretary, Student Gymkhana, IIT Mandi

Jul 2026 – Present

- Elected student lead for academics for ~**3,500 undergraduates**; head the UG Academic Council; selected and oriented **12 branch councillors**.
- Ran Batch 2026 academic orientation and **15 branch-wise sessions**; prepared proposals for the 65th and 66th Board of Academics, including a **388-page** curriculum booklet.
- Proposed an institute-hosted LaTeX platform backed by a **483-response** survey; audited the Academic Section website with **34 changes** adding 568 approved courses.
- Organised Teachers' Day 2026 (chief guest Prof. Prem Kumar Kalra), reaching more than 60% of faculty.

Batch 2023 Academic Representative & Academic Guide, UG Academic Council

2024 – 2026

- Liaised between **500+ students** & faculty; co-presented the first post-COVID offline Curriculum Explanation Session; hired the UGAC web team (8 candidates, 4 selected); organised *VyarthShastra*.

Head of Technical Team, Exodia 2026 (Annual Cultural Fest)

Apr 2026

- Owned fest tech: solo-built the ticketing and gate system and tuned exodia26.org performance.

Instructor, HIMSHIKHAR (CCE IIT Mandi × Masai) & PRAYAS 4.0

Jun 2026 – Jul 2026

- Taught **22 classes (40+ hours)**, DBMS to GenAI, and an 18-hour ML course with 4 Colab labs; led **8 TAs**.

UG Head Teaching Assistant, Machine Learning

- Supervised course operations for **600+ students** across a full semester.

Mentoring & Clubs

2023 – Present

- GCS Cell Mentor; SAE Club Mentor; Core Member of STAC (hackathons for **100+ teams**); JEE Mentor at Physics Wallah (**100+** aspirants).

ACHIEVEMENTS

AIR 1, Global Rank 11, SAND 2025 (IEEE ICASSP 2026 challenge): ALS severity from voice; India #2, Global 19 in Task 2.

Rank 10, solo vs teams, Vehant Vehicle Count Challenge: classical CV only, MAE 16.4 (Jan 2026).

Finalist, Pune International Agri Hackathon 2026: Geospatial Crop Survey track (May 2026).

3rd Rank, AiHack India Hackathon 2025: hosted at IIT Mandi by Aiful Corporation, Japan (Nov 2025).

Round 2 (top 1.5%), Adobe India Hackathon 2025: team leader among 262,120 teams (Jul 2025).

Emerald of Gymkhana Award: best GCS Cell volunteer, Dean Students' Office (May 2025).

National Semi-Finalist, Tata Imagination Challenge 2024: through 2 national rounds (Feb 2025).

1st Rank, Space Code Challenge 2023: STAC, IIT Mandi (Jan 2024).

20+ hackathons & data challenges in 2025–26, incl. Amazon ML Challenge 2025 (82,787 teams), IDBI Innovate 2026 (team lead), ABB Accelerator 2026, SEBI GFF 2025.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C, C++, Java, Kotlin, SQL, Rust (Tauri), R, HTML/CSS, LaTeX

Backend: FastAPI, Flask, Django, Node.js, Express, Fastify, Hono, tRPC, REST, OpenAPI, WebSockets, Socket.IO, Kafka

Frontend & Apps: React, Next.js, Vite, Tailwind, Astro, Tauri, React Native (Expo), Android (Kotlin), PWAs

Data: PostgreSQL, SQLite, MongoDB, Redis, Prisma, Drizzle, Firestore, FAISS

ML & AI: PyTorch, TensorFlow/Keras, scikit-learn, LightGBM, XGBoost, OpenCV, Transformers, LoRA/PEFT, CLIP, DINOv2, SAM2, LangChain, RAG, MLflow

Cloud & DevOps: Docker, AWS (EC2, RDS, S3), DigitalOcean, Vercel, Render, Firebase, nginx, pm2, Caddy, CI/CD, Playwright, Linux, Raspberry Pi

Integrations: Razorpay (Route, RazorpayX), Zoho Books, WhatsApp Cloud API, Keycloak SSO, NextAuth/OAuth